

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	05 AUG 2026
Student Name	P. Srujan Sinha
Project Name	Intergalactic Space Travel Request (ISTR) System
Maximum Marks	4 Marks

Functional Requirements

The **Intergalactic Space Travel Request System (ISTR)** provides a comprehensive set of functional requirements to automate the complete lifecycle of space mission requests. These requirements define the core features of the application, ensuring efficient request submission, multi-level approval processing, workflow automation, mission tracking, reporting, and secure user management.

FR No.	Functional Requirement (Epic)	Description	Sub Requirements (Stories / Tasks)
FR-1	Mission Request Management	Enables users to create, manage, and maintain mission requests throughout their lifecycle.	• Create Mission Request • View Mission Request • Edit Mission Request • Cancel Mission Request
FR-2	Approval Workflow	Implements a multi-level approval process to ensure mission requests are reviewed and authorized by the appropriate stakeholders.	• Manager Approval • Finance Approval • Mission Control Approval
FR-3	Workflow Automation	Automates mission processing using ServiceNow Flow Designer, reducing manual intervention and improving operational efficiency.	• Automatic Status Updates • Email Notifications • Flow Designer Automation
FR-4	User & Access Management	Controls user authentication, authorization, and secure access to application resources based on assigned roles.	• Role-Based Access Control (ACLs) • User Groups & Roles • Secure Authentication
FR-5	Mission Tracking & Reporting	Provides real-time visibility into mission progress through dashboards, reports, and analytical insights.	• Dashboard Monitoring • Mission Status Tracking • Reports & Analytics
FR-6	Form Validation & Business Logic	Ensures data accuracy and enforces business policies using platform-level validations and automation.	• Client Script Validation • UI Policies • Business Rules

Non-Functional Requirements

The **Non-Functional Requirements (NFRs)** define the quality attributes of the Intergalactic Space Travel Request System (ISTR). These requirements ensure that the application remains secure, reliable, scalable, efficient, and easy to use while delivering a consistent user experience.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system shall provide an intuitive, responsive, and user-friendly interface through the Service Catalog and Service Portal, enabling users to submit, monitor, and manage mission requests with minimal training.
NFR-2	Security	The application shall implement Role-Based Access Control (ACLs), secure authentication, authorization mechanisms, and data protection policies to ensure confidentiality and integrity of mission information.
NFR-3	Reliability	The system shall ensure accurate workflow execution, consistent data processing, dependable approval routing, and reliable notification delivery throughout the mission lifecycle.
NFR-4	Performance	Mission requests, approvals, reports, dashboards, and notifications shall be processed with minimal response time while maintaining optimized workflow execution under normal operating conditions.
NFR-5	Availability	The application shall remain continuously available to authorized users with high system uptime, ensuring uninterrupted access to mission management and approval services.
NFR-6	Scalability	The system shall efficiently support increasing numbers of users, mission requests, workflows, approvals, and reports without significant degradation in performance or responsiveness.